One mean and variance per channel

each image: C = 3 channel sheets, channel c in front

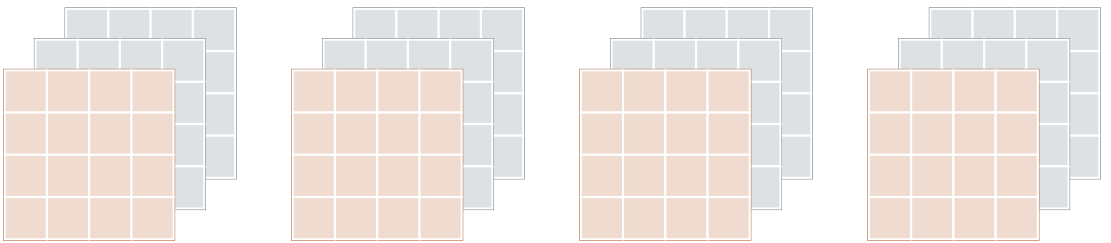

image 1 image 2 image 3 image 4

channel c across the batch and positions: B × H × W values

$$\hat{x} = (x - \mu_c) / \sqrt{\sigma_c^2 + \varepsilon}, \quad y = \gamma_c \hat{x} + \beta_c$$

BatchNorm2d(C): 2C learned (γ, β) + 2C running statistics